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Review Article

Cognitive functioning in children with both epilepsy and ADHD: A systematic review and meta-analysis

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ABSTRACT

There is a well-known bidirectional association between epilepsy and attention deficit hyperactivity disorder (ADHD) in children. The aim of the current systematic review and meta-analysis was to investigate neuropsychological functioning of children with epilepsy (CWE) and ADHD. The systematic search of two databases revealed 22 observational studies that comprised samples of CWE + ADHD and cognitive outcomes as compared to CWE alone, ADHD alone, and/or healthy controls. Neuropsychological data was organised with respect to Cattell-Horn-Carroll (CHC) taxonomy. Random effects meta-analyses across all measured cognitive domains showed small to moderate effect sizes indicating poorer cognitive outcomes in the CWE + ADHD group as compared to either condition alone (standardised mean difference [SMD] -0.30 to -0.72), with moderate to large effect sizes compared to healthy controls (SMD -0.61 to -1.12). Confidence intervals were wide and overlapping, suggesting a broad heterogenous profile of neuropsychological functioning in children with both conditions. These findings underscore the breadth, magnitude and vulnerability of cognitive dysfunction in children with comorbid neurobehavioral conditions, thus highlighting the need for comprehensive assessment and targeted intervention in this vulnerable population.

1. Introduction

Epilepsy and attention deficit hyperactivity disorder (ADHD) represent the most common neurological and neurodevelopmental disorders in children, affecting up to 1% and 5–7% of the population, respectively [1–2]. There is significant comorbidity among children with epilepsy (CWE) and ADHD, with estimates of up to 30% of CWE meeting diagnostic criteria for ADHD [3] with a higher prevalence inattention (rather than hyperactivity/impulsivity) symptoms [4], and a 2.5-fold higher risk of epilepsy in those with ADHD [5].

Understanding the nature of the combined presentation is complex, and considers shared context at the genetic, biological, and environmental levels interacting over time [6]. For instance, there are shared genetic factors that may link the conditions which, when interacting with environmental variables may modify developmental trajectories and behavioural vulnerabilities [7]. Complexity increases when there is consideration to the many biological variables that contribute to and/or exacerbate the cognitive and behavioural challenges (e.g., medication effects, impact of seizures, interictal paroxysmal activity etc.; [8]). Research has shown that in many children with both ADHD and epilepsy, challenges with learning, cognition and socio-emotional function antedate the onset of seizures and have been linked with structural brain differences [7,9]. A recent meta-analysis of prevalence studies showed that age (e.g., younger, earlier age of onset), epilepsy type (e.g., absence epilepsy), frequency of seizures and intellectual/developmental disabilities were significant risk factors for ADHD in epilepsy [3].

There is an extensive body of literature on cognitive outcomes in both childhood epilepsy and ADHD, although historically, there has been a focus on understanding specific profiles of neuropsychological function at the group level across different epileptic syndromes [10], and typical domains of dysfunction in ADHD (e.g., attention and executive functioning; [11]). The review literature has revealed significant heterogeneity, and widespread deficits present at the group study level [12–16]. Considering this variability, there has been a movement towards understanding broad cognitive phenotypes within a personalised medicine approach and a paradigm shift in the understanding of neuropsychological outcomes and neurobehavioral comorbidities moving towards network-based models of brain function [10,17]. As Hermann et al. [10] highlight, epilepsy is increasingly viewed as a disorder

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involving widespread disruptions in brain networks, extending beyond the regions responsible for primary seizure generation, and therefore considering the dysfunction of related networks can also extend beyond seizures themselves. Others have considered the utility of classifying a broader overarching developmental brain disorder with varying topography, as opposed to the consideration of discrete comorbid conditions [18].

Research has documented a vulnerability to more severe cognitive deficits in the comorbid ADHD/epilepsy presentation [19] although to date, there has been no effort to quantitively synthesize the available research to determine the magnitude and profile of neuropsychological outcomes. The recent systematic review by Leal and colleagues [19] acknowledged the similarities of their review to the present pre-registered review, however they focussed primarily on exploring how ADHD alters neurocognitive outcomes in CWE. While acknowledging the heterogeneity between studies that is often present when considering such populations broadly, information from quantitative syntheses may serve to further clarify the comorbid relationship and associated combinatorial and/or overlapping cognitive features. The aim of the current systematic review and meta-analysis was to synthesize current research on neuropsychological functioning of children with both epilepsy and ADHD, according to a well-defined taxonomy of cognition, and to compare these profiles with those of CWE alone, ADHD alone, and/or healthy controls.

2. Methods

2.1. Search strategy

The systematic review protocol was registered with the International Prospective Register of Systematic Reviews (PROSPERO: registration number CRD420251015603). The review was conducted according to PRISMA guidelines [20]. A search for articles was conducted using Pubmed and ProQuest databases to May 2025. Keywords were entered in terms of epilepsy (epilepsy OR seizures), attention deficit hyperactivity disorder (ADHD), and cognitive function (*cognitive, cognition, neuropsychology, neuropsychological, intelligence, language, memory, executive function*). The reference lists of eligible articles were also searched for additional studies. Further detail regarding search strategy, PRISMA checklist and protocol registration can be found in the supplementary material.

2.2. Selection criteria

Study inclusion criteria were all of the following: (a) original research published in a peer-reviewed journal, (b) studies with samples including participants with both childhood epilepsy and ADHD in addition to a control group with either healthy peers, epilepsy without ADHD or ADHD without epilepsy (c) population included children up to 16 years of age (d) any published measure of neuropsychological functioning.

Studies were excluded if (a) separate data for a combined epilepsy and ADHD group was not provided (b) epilepsy was in the context of another defined genetic or neurological condition (e.g., stroke, brain injury, trisomy 21 etc.), (c) the control group was absent or some other diagnosable condition; (d) full-text article was not available; (e) means and standard deviations for test data were not available.

2.3. Data Extraction

Data extraction was completed by corresponding author, SW. Means and standard deviations from measures of cognitive function were extracted. Based on the methodology of previous systematic reviews in epilepsy [13–14,21], the Cattell-Horn-Carroll (CHC) model was used to classify broad cognitive factors for each test. The CHC model of intelligence is widely regarded as the most comprehensive and empirically supported framework for understanding human cognition [22]. It is structured as a hierarchical, three-stratum model. At the top level is a general intelligence factor (G), which influences a range of broad cognitive abilities and each of these broad domains encompasses numerous more specific, narrow abilities. For the purpose of the present study, the relevant broad abilities included:

- **Crystallized intelligence (G*c*)** - *acquired knowledge of the language, information and concepts of a specific culture, and/or the application of this knowledge.*
- **Fluid reasoning (G*f*)** - *deliberate and controlled mental operations to solve novel problems that cannot be performed automatically.*
- **Short-term Working memory (G*sm*)** – *limited capacity system for holding and working with information in the immediate situation.*
- **Processing speed (G*s*)** - *the ability to automatically and fluently perform relatively easy or over-learned elementary cognitive tasks.*
- **Decision Speed/Reaction Time (G*t*)** - *the ability to make elementary decisions and/or responses at the onset of simple stimuli.*
- **Long-term storage and retrieval and fluency (G*lr* / G*r*)** - *the ability to store and consolidate new information in long-term memory and later fluently retrieve the stored information.*

Regarding neuropsychological test extraction of data, where a single study reported on multiple tests reflecting the same underlying broad CHC factor (i.e. PIQ and Raven's matrices*)* each test was extracted and pooled within the study for the respective factor (I.e. *Gf)*. Where a test included multiple outcomes for the same set of stimuli, a single representative outcome was chosen (i.e. reaction time variability in continuous performance tasks). Where separate neuropsychological test data was reported for subgroups within a study (i.e. ADHD or epilepsy subtype), data for each group was extracted and pooled within the study. Where information was reported prospectively across multiple timepoints, data was extracted for patients and controls at T0 only (e.g. [9]). Where there were multiple papers containing possible overlapping samples (i.e. [7,9]) papers/outcomes were retained if they provided information on different cognitive measures and/or domains of function. There was one full text paper by Fang et al. [23] available in Chinese and unavailable in English, which was included and translated to English using Google Translate.

Classifying tests traditionally conceptualised as measures of executive function is challenging within the CHC taxonomy, as many measures overlap and are subsumed under multiple CHC constructs, including but not limited to Gf, Gs, Gsm, and Gr. Accordingly, tests were categorised based on the factor judged to account for the greatest proportion of variance in the measurement outcome. For example, executive measures of inhibitory control, such as the Stroop task, were classified under processing speed (Gs), as they are considered measures of complex speed due to their primary reliance on response time. This approach has been adopted in previous research [13–14].

In addition to cognitive outcomes, information was extracted from included studies pertaining to study characteristics, demographics and key clinical variables. This included case and control sample sizes; country of study; mean sample age of groups; mean sample age of onset for epilepsy; epilepsy type (e.g., focal/generalised/epileptic syndrome); ADHD and epilepsy diagnostics; and medication status of clinical groups (e.g., antiepileptics and/or stimulants). Meta-regression and/or moderator analyses were considered if there were a sufficient number of studies ($k \geq 10$).

2.4. Statistical analysis

Meta-analyses were performed using the Metafor package in RStudio 4.3.1. Effect sizes were calculated for each study and then pooled by cognitive domain. Each study contributed an effect size representing the standardised mean difference (SMD) in neuropsychological performance within the domain between children with both ADHD and epilepsy, and

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those with either condition alone or healthy controls. Standard benchmarks were used to interpret effect size magnitude (SMD = 0.20 = small, 0.50 = moderate, 0.80 = large), and interpretations were supported by the corresponding confidence intervals. A random-effects meta-analysis was conducted for each cognitive domain, estimating the overall mean effect size and its 95% confidence and prediction interval. This model assumes that the included studies are independent, and each reflects a true effect size drawn from a normally distributed population. Heterogeneity among studies was assessed using the Q test, where a significant result indicates variability beyond chance. The I^2 statistic was also calculated to quantify the proportion of variance due to heterogeneity rather than random sampling error.

2.5. Assessment of methodological quality

Methodological quality of studies was completed using the Newcastle Ottawa Scale (NOS) for case-control studies [24]. The NOS scale assesses each study for risk of bias across three domains of selection, comparability and exposure/outcome, with a total possible score out of nine. Scores are classified as low risk (i.e. 7–9), moderate risk (i.e. 4–6), and high risk (i.e. 0–3) of bias. In the current context of observational, cross sectional neuropsychological outcome data between cases (CWE + ADHD) and controls (CWE, ADHD or healthy controls), all studies were rated according to case-control NOS criteria, as the distinction between case-control and cohort designs was not considered relevant to the current study. It is acknowledged that the registered protocol specified independent quality assessment by two assessors, in practice a second assessor (PG) independently evaluated a randomly selected 30% of studies (7/22) to assess inter-rater reliability (Cohen's κ). Where there were discrepancies between raters, consensus was reached through collaborative discussion.

3. Results

Retrieved searches were imported into Rayyan for screening and duplicate search [44]. The initial search identified 1361 records and there were 730 studies after duplicate removal. Among these, 99 full-text papers reviewed for eligibility. Exclusion of 77 studies resulted in 22 studies that met the inclusion criteria (see Fig. 1 for the Prisma Flowchart and Table 1 for characteristics of included studies). Included studies had samples from varied geography, although approximately a third of samples were US based. The average age of study CWE + ADHD samples ranged from 7.6 to 12.1 years old. Studies comprised samples with mixed epilepsy syndromes (57%), focal (30%) or generalised epilepsy (13%). Across the various cognitive domains, the number of included studies ranged from 3 to 13. For the combined group, the sum number of cases ranged from 50 to 405 children and comparison groups from 77 to 717 children.

The G factor comprised tests largely reflective of Full-Scale Intelligence Quotient from various editions of the Wechsler battery (i.e., WISC-R, WISC-III, WISC-IV, WASI) across different language-editions. Fluid reasoning (Gf) measures were represented by non-verbal performance and perceptual reasoning Wechsler tests, along with other flexible thinking and mental switching tasks that are often considered under the umbrella of executive functioning (e.g., Ravens' Matrices, Wisconsin Card Sorting Test, Mazes, and Tower of Hanoi). The domain of crystallized intelligence (Gc) was represented by the verbal comprehension composites of the Wechsler scale (i.e. VIQ and VCI), as well as language

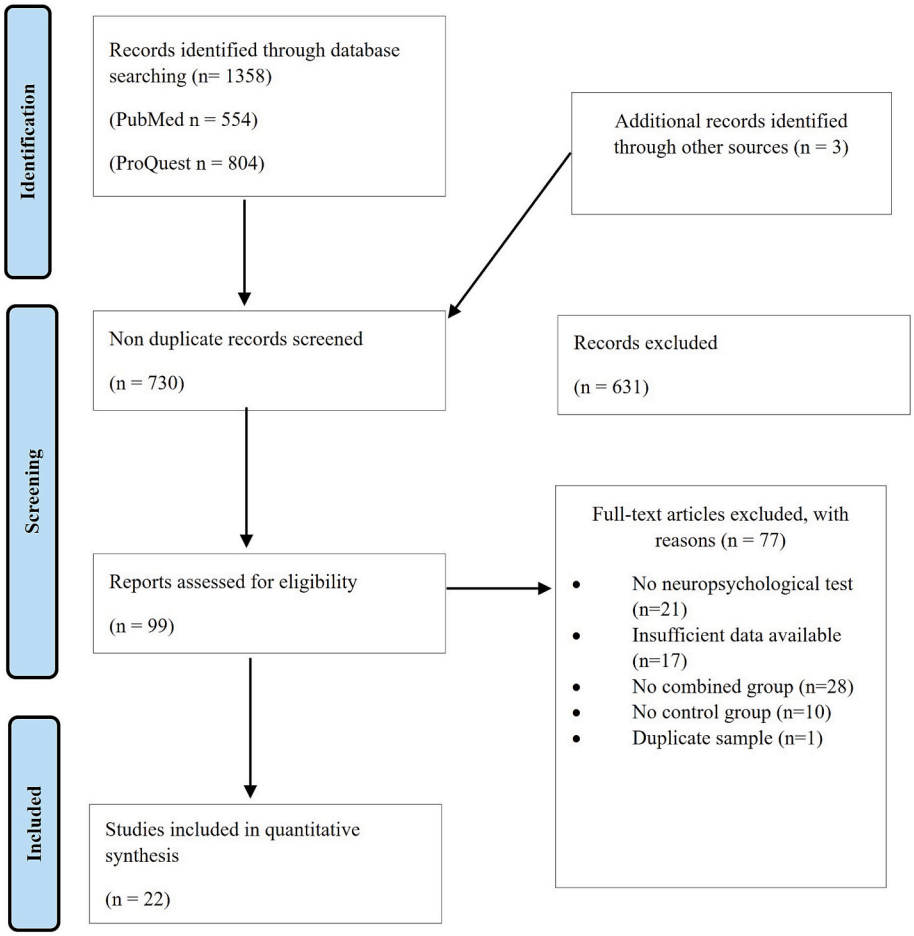


Fig. 1. Prisma flow chart of systematic search.

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Table 1
Characteristics of included studies.

Study	Country	Group (n)	Age	Epilepsy			ADHD				
			M (SD)	Epilepsy Type (F/G/C)	Age at onset (m [sd])	Med (% on AED/s)	Dx criteria/tool	Type (I/H/C)	Med	IQ Exclusion	ROB (NOS)
Ahmed et al. [25]	Egypt	CWE + ADHD (20) HC (20)	7.6 (1.9) 7.8 (1.4)	0/20/0	2.75 (1.2)	80%	DSM-5 CPRS	−/−/10	N/R	<70	High (3)
Bakhtadze et al. [26]	Israel	CWE + ADHD (25) ADHD (18) HC (26)	10.16 (0.28)[1] 9.61 (0.14) 8.26 (0.26)	TLE	N/R	N/R	DSM-IV CPRS	N/R	N/R	<70	Moderate (4)
Bechtel et al. [27]	Germany Switzerland	CWE + ADHD (17) ADHD (15) HC (15)	11.5 (1.6) 11 (1.4) 11.5 (1.8)	12/2/3	N/R	82%	DSM-IV CPRS	6/9	53%	<70	High (3)
Choudhary et al. [28]	India	CWE + ADHD (17) CWE (56)	8.52 (1.68) 8.63 (1.84)	9/8/0	2.88 (2.45)	100%	DSM-IV TR CBCL	10/1/6	N/R	<70	Moderate (5)
Costa et al. [29]	Brazil	CWE + ADHD (6) CWE (30) HC (37)	11.2 (2.1) 11.4 (2.1)	5/5/0	N/R	94%	DSM-IV SNAP-IV	3/0/3	N/R	<80	Moderate (7)
Fang et al. [23]	China	CWE + ADHD (80) CWE (91) ADHD (70) HC (70)	7.9 (1.8) 8.4 (1.8) 8.2 (1.4) 8.2 (1.6)	SeLECTS	N/R	N/R	DSM-5	N/R	N/R	Severe ID	Moderate (5)
Gómez-Castrillón et al. [30]	Columbia	CWE + ADHD (20) ADHD (20)	8.9 (1.6) 9.3 (2.1)	SeLECTS	N/R	100%	DSM-5	0/0/20	N/R	<80 (RIAS)	Moderate (4)
Hermann et al. [9]	US	CWE + ADHD (15) CWE (24) HC (38)	N/R N/R 12.7 (3)	19/9/0[2]	10.9 (3.8)	N/R	DSM-IV K-SADS	NR	N/R	N/A	Low (9)
Hermann et al. [7]	US	CWE + ADHD (23) CWE (52) HC (58)	11.6 (3.2) 12.9 (3.2) 13 (3.1)	14/9/0	10.98 (3.1)	91%	DSM-IV K-SADS	12/4/3/4	43%	N/A	Low (9)
Huang et al. [31]	China	CWE + ADHD (29) CWE (89)	8.6 8.3	SeLECTS	~80% between 5-10yo	100%	DSM-5	N/R	N/R	N/A	Moderate (5)
Kinney et al. [32]	US	CWE + ADHD (16) ADHD (85)	10.2 (2.5) 10 (3.0)	N/R	N/R	N/R	DSM-III (best estimate method)	N/R	8%	Severe ID	High (3)
Kral et al. [33]	US	CWE + ADHD (132) CWE (72)	10.62 (3.27)[1] 12.33 (3.54)	68/34/?30	4.88 (3.84)	N/R	DSM-IV-TR BASC-2	60/0/72	N/R	N/A	Moderate (4)
Lee et al. [34]	US	CWE + ADHD (23) CWE (42) ADHD (42) HC (42)	10.8 (2.7) 10.9 (2.5) 10 (2.2) 10.1 (2.4)	23/0/0	N/R	100%	DSM-IV Conners P&T	13/0/10	N/R	N/A	Moderate (6)
Lee et al. [35]	Korea	CWE + ADHD (8) CWE (12)	7.2 (1.6) 7.9 (1.8) 8.2 (1.8)	CAE	N/R	Nil	DSM-IV ADHD-RS	N/R	Nil	<70	Moderate (4)

(*continued on next page*)

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Table 1 (*continued*)

Study	Country	Group (n)	Age	Epilepsy			ADHD				
			M (SD)	Epilepsy Type (F/G/C)	Age at onset (m [sd])	Med (% on AED/s)	Dx criteria/tool	Type (I/H/C)	Med	IQ Exclusion	ROB (NOS)
		ADHD (20) HC (20)									
Lee et al. [36]	Korea	CWE + ADHD (13) CWE (78)	10.1 (median)	Mixed sample	9.7 (2.9)	Nil	ADS ADHD-RS	N/R	Nil	<50	Moderate (4)
Lima et al. [37]	Brazil	CWE + ADHD (12) CWE (11) ADHD (20) HC (20)	9.7 (2.4) 11.9 (1.8) 10.9 (2.5) 10.2 (2.5)	SeLECTS	7 (2.9) (ADHD+CWE and CWE)	74% (ADHD+CWE and CWE)	DSM-IV K-SADS	9/1/2	Nil	<80	Moderate (5)
MaCallister et al. [38]	US	CWE + ADHD (46) ADHD (45)	10.8 (3.4) 11.9 (2.8)	25/17/4	6.13 (3.65) /4.3 (2.69) (I-ADHD, C-ADHD)	84%	American Academy of Pediatrics	31/0/15	N/R	Nil ID	Moderate (4)
Park et al. [39]	Korea	CWE + ADHD (33) CWE (113) ADHD (294)	7.9 (2) 8.2 (2.3) 7.7 (1.7)	14/18/1	N/R	Nil	DSM-IV-TR ADHD-RS	13/2/8	Nil	ID (<70)	Moderate (5)
Semrud-Clikeman et al. [40]	US	CWE + ADHD (12) CWE (21) ADHD (22) HC (15)	9.9 (3) 11.9 (3.7) 10.2 (3.2) 10.3 (3.1)	0/12/0	7/12 < 7yo	100%	DSM-III R K-SADS	N/R	100%	<70	Moderate (4)
Thieux et al. [41]	France	CWE + ADHD (67) CWE (55) ADHD (52) HC (53)	10.3 (3.3) 10.9 (3.3) 11.1 (2.1) 11.6 (3.5)	26/41	5.33 (3.42)	69%	DSM-V ADHD-RS	N/R	Nil	N/A	Moderate (6)
Tsai et al. [42]	Taiwan	CWE + ADHD (15) CWE (46)	9.8 (2.5)[3]	Mixed sample	3.35 (2.66)	100%	DSM-IV SNAP-IV	7/3/5	Nil	<70	High (3)
Xaio et al. [43]	China	CWE + ADHD (15) CWE (15) HC (15)	8.2 (1.9) 8.7 (2.2) 8.4 (2.1)	SeLECTS	7.5 (1.7)	100%	DSM-IV ADHD-RS	N/R	Nil	<75	Low (7)

n = sample size, M = mean, SD = standard deviation, Epilepsy type (F/G/C) = focal/generalised/combined, Med = Medication, AED = anti-epileptic drug, Dx = diagnostic, ADHD = Attention Deficit Hyperactivity Disorder, ADHD type (I/H/C) = inattentive/hyperactive/impulsive/combined, IQ = intelligence quotient, ID = Intellectual Disability, NOS = Newcastle-Ottawa Scale, CWE = Children With Epilepsy, HC = Healthy Controls, N/R = not reported, N/A = not applicable, TLE = Temporal Lobe Epilepsy, CAE = Childhood Absence Epilepsy, SeLECTS=Self Limited Epilepsy with Centrotemporal Spikes, ROB = Risk of Bias, DSM = Diagnostic and Statistical Manual, CPRS = Conners Parent Rating Scale, CBCL = Child Behaviour Checklist, SNAP=Swanson, Nolan and Pelham Rating Scale, ADS = ADHD Diagnostic System, RIAS = Reynolds Intelligence Scale, K-SADS = Kiddie Schedule for Affective Disorders and Schizophrenia, BASC =Behaviour Assessment System for Children, ADHD-RS = Attention Deficit Hyperactivity Disorder Rating Scale [1] = age was averaged for the ADHD+CWE from the provided subgroups, [2] = taken from the CWE-Comorbidity+ group (ADHD and Academic Problems), [3] = average age of both CWE groups.

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measures such as the Boston Naming Test and expressive vocabulary tests. Studies examining the short-term working memory construct (Gsm) employed tasks from the Wechsler working memory tasks, or memory batteries (i.e., Wide Range Assessment of Learning and Memory [WRAML], Children Memory Scales [CMS]), examining both verbal and visuospatial short-term working memory abilities. The broad stratum ability of processing speed (Gs) contained standard tasks of speed of information processing of simple visual material, such as the PSI measures from the Wechsler scales, Trail Making and Stroop tasks. Reaction and decision time (Gt) comprised tests representing sustained attention and vigilance measures from Continuous Performance Tasks (CPTs), principally attending to measures of Reaction Time Variability. When only accuracy measures of continuous performance tasks were available, these were considered more compatible with Gs over Gt. Given paucity of studies including measures of long-term memory retrieval and fluency independent of encoding, both verbal learning tasks and phonemic and semantic fluency tasks were pooled into a representative G*lr* construct. A detailed list of neuropsychological tests organised according to broad stratum CHC constructs can be found in the supplementary material.

3.1. Quantitative synthesis

Summary statistics for random effects meta-analyses of cognitive outcomes in the combined CWE + ADHD group as compared to either group alone (i.e. CWE or ADHD alone) and healthy controls are demonstrated in Table 2. Pooled standardised mean differences (SMD) with confidence intervals across each cognitive domain and compassion group are visualised in Fig. 2. Compared to children with either ADHD or epilepsy alone, there were small to moderate effect sizes observed across all assessed domains of cognition with overlapping confidence intervals, indicating poorer outcomes in the CWE + ADHD group.

In the ADHD alone comparison, the largest effect size was in the Glr domain (k = 3, SMD = −0.72, 95% CI = −0.94 to −0.50, $p < 0.001$), however this also contained the fewest studies. Smaller effects were seen in domains of short-term working memory (k = 6, SMD = −0.31, 95% CI = −0.59 to −0.03, $p = 0.03$) and reaction time / attention measures from continuous performance tasks (k = 5, SMD = −0.33, 95% CI = −0.84 to −0.17, $p = 0.017$) followed by the overarching factor G (k = 7, SMD = −0.44, 95% CI = −0.70 to −0.17, $p = 0.001$) and fluid intelligence (k = 8, SMD = −0.38, 95% CI = −0.60 to −0.15, p = 0.001). There were relatively larger, moderate effects noted in verbal language abilities (k = 7, SMD = −0.54, 95% CI = −0.85 to −0.23, $p < 0.001$) and processing speed (k = 5, SMD = −0.56, 95% CI = −0.94 to −0.17, $p = 0.005$).

Random effects meta-analyses comparing the ADHD+CWE group to CWE alone revealed mostly small effect sizes across domains of Gsm (k = 5, SMD = −0.30, 95% CI = −0.47 to −0.12, p < 0.001), processing speed (k = 6, SMD = −0.32, 95% CI = −0.49 to −0.15, $p < 0.001$) and reaction time / decision speed (k = 6, SMD = −0.45, 95% CI = −0.84 to −0.06, $p = 0.024$), Glr (k = 3, SMD = −0.34, 95% CI = −0.55 to −0.12, $p = 0.002$), fluid intelligence (k = 8, SMD = −0.44, 95% CI = −0.76 to −0.12, $p = 0.008$) and the overarching G factor (k = 11, SMD = −0.43, 95% CI = −0.68 to −0.18, $p < 0.001$). There was a moderate effect size for Gc (k = 10, SMD = −0.55, 95% CI = −0.85 to −0.25, p < 0.001).

Alternatively, the CWE + ADHD group showed large effect sizes compared to the healthy control group across most cognitive domains (k = 4–8, SMD = −0.70 to −1.12, $p < 0.01$) except for Glr, where there was an observed moderate effect size (k = 3, SMD = −0.61, 95% CI = −0.82 to −0.41, $p < 0.001$). Moreover, the effect for Gt compared to healthy controls did not reach significance ($p = 0.053$).

3.2. Clinical variables and neuropsychological function

Among the various analyses, there were two cognitive domains in the CWE control group where there were sufficient studies (i.e. ≥ 10) for performing meta-regression and subgroup analyses (i.e., G and Gc). Meta-regression was considered for the following variables: age, age of onset, epilepsy subtype (e.g., % of sample focal), AED status, ADHD subtype and NOS quality rating. Overall, no tested variable explained any significant portion of variance, however, there were several missing values across studies severely limiting the analyses.

When consulting included study effects across epilepsy variables there were inconsistent effects on cognitive outcomes. An earlier age of onset and a higher number of antiepileptic drugs (AEDs) was associated with poorer attention in CWE, both with and without ADHD in one study [41]. However, several other studies failed to show any significant difference in cognitive measures based on age of seizure onset or AED usage [9,29,36,38]. While some studies did not reveal any significant differences between neuropsychological outcomes and epilepsy type, duration, and seizure frequency [9,36,41] one study found that individuals

Table 2
Summary results of random effects meta-analyses for cognitive outcomes in combined CWE + ADHD.

Control Group	Outcome	Studies, k	CWE + ADHD, n	Controls, n	SMD	*P*-value	95% CIs	95% PIs	I2	Tau2	Q test p-value
ADHD	G	7	219	566	−0.44	0.001	−0.70, −0.17	−0.99, 0.12	51.17%	0.25	0.06
	G*f*	8	226	319	−0.38	0.001	−0.60, −0.15	−0.85, 0.09	43.57%	0.21	0.10
	G*c*	7	209	314	−0.54	< 0.001	−0.85, −0.23	−1.24, 0.17	62.83%	0.33	0.01
	G*sm*	6	198	212	−0.31	0.030	−0.59, −0.03	−0.94, 0.32	71.88%	0.29	0.01
	G*s*	5	160	196	−0.56	0.005	−0.94, −0.17	−1.38, 0.28	74.13%	0.38	0.01
	G*t*	5	166	431	−0.33	0.017	−0.61, −0.06	−0.84, 0.17	50.39%	0.21	0.08
	G*lr*	3	55	82	−0.72	< 0.001	−0.94, −0.50	−0.94, −0.50	0.00%	0	0.99
CWE	G	11	370	625	−0.43	< 0.001	−0.68. -0.18	−1.10, 0.25	63.02%	0.32	0.002
	G*f*	8	190	386	−0.44	0.008	−0.76, −0.12	−1.26, 0.38	71.13%	0.38	0.001
	G*c*	10	222	457	−0.55	< 0.001	−0.85, −0.25	−1.38, 0.28	70.64%	0.39	0.001
	G*sm*	5	53	246	−0.30	< 0.001	−0.47, −0.12	−0.55, −0.04	21.85%	0.01	0.33
	G*s*	6	268	270	−0.32	< 0.001	−0.49, −0.15	−0.58., −0.06	21.84%	0.01	0.38
	G*t*	6	258	293	−0.45	0.024	−0.84, −0.06	−1.37, 0.46	80.41%	0.18	<0.001
	G*lr*	3	50	77	−0.34	0.002	−0.55, −0.12	−0.55, −0.12	37.00%	<0.001	0.34
HC	G	7	166	261	−1.06	0.001	−1.71, −0.41	−2.77, 0.64	86.43%	0.86	<0.0001
	G*f*	8	185	257	−1.12	< 0.001	−1.73, −0.51	−2.86, 0.62	89.80%	0.83	<0.0001
	G*c*	7	163	237	−0.94	0.008	−1.64, −0.24	−2.82, 0.93	91.58%	0.79	<0.0001
	G*sm*	5	138	184	−0.89	0.012	−1.59. -0.19	−2.52, 0.74	92.69%	0.56	<0.0001
	G*s*	6	161	233	−1.03	0.002	−1.66, 0.39	−2.63, 0.58	91.89%	0.57	<0.0001
	G*t*	4	93	130	−0.70	0.053	−1.41, 0.01	−2.19, 0.78	85.33%	0.44	<0.001
	G*lr*	3	50	100	−0.61	< 0.001	−0.82, −0.41	−0.82, −0.41	0.00%	0	0.69

ADHD = Attention Deficit Hyperactivity Disorder, CWE = Children With Epilepsy, HC=Healthy Controls. K = number of studies, n = sample size, SMD=Standardised Mean Difference, CI=Confidence Interval, PI=Prediction Interval, G = General intelligence factor, Gc = Crystallized intelligence, Gf = Fluid reasoning, Gsm = Short-term working memory, Gs = Processing speed, Gt = Decision speed/reaction time, Glr = Long-term storage and retrieval, and fluency.

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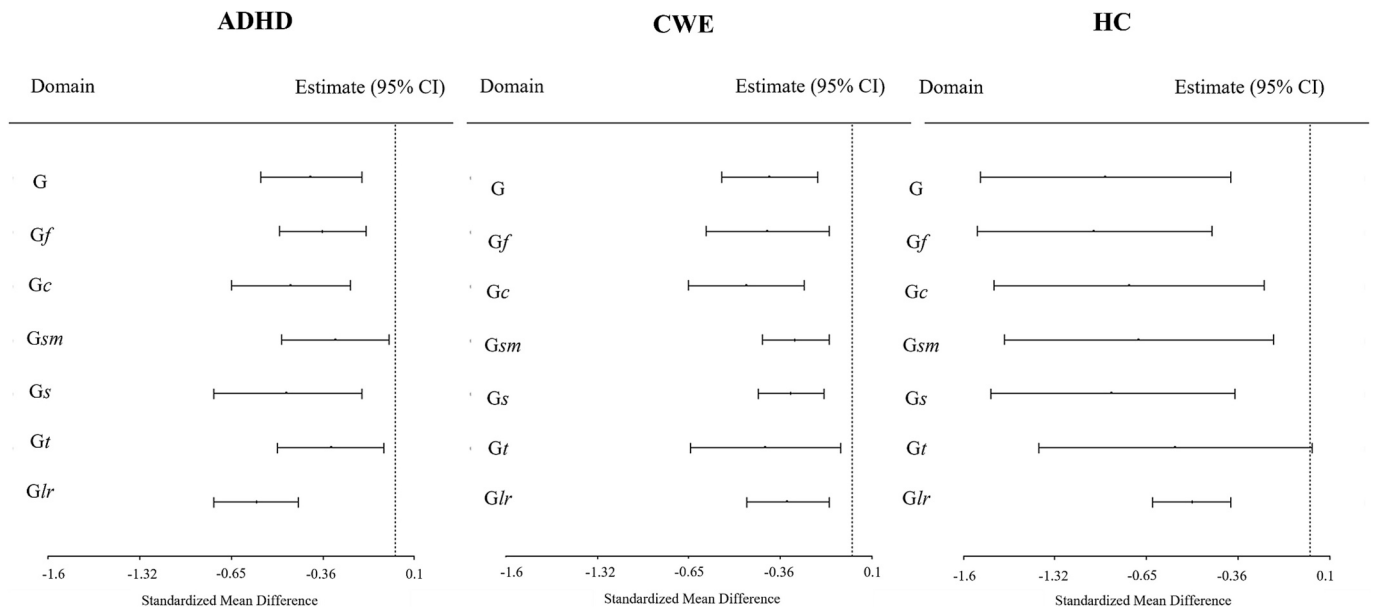


Fig. 2. Summary forest plots of CWE + ADHD compared to ADHD, CWE and healthy controls, ADHD = Attention Deficit Hyperactivity Disorder, CWE = Children With Epilepsy, HC = Healthy Controls, CI = Confidence Interval, G = General intelligence factor, Gc = Crystallized intelligence, Gf = Fluid reasoning, Gsm = Short-term working memory, Gs = Processing speed, Gt = Decision speed/reaction time, Glr = Long-term storage and retrieval, and fluency.

with idiopathic generalised epilepsy (IGE) had significantly lower FSIQ and CPT omission errors compared to those with idiopathic focal epilepsy, but not SeLECTS [36]. Some specific associations were shown by Costa et al. [29] regarding seizure variables and attention, such as between focal seizures and sustained attention, as well as more seizures impacting alternating attention. Seizure frequency was linked with lower full-scale IQ and discrepancies between perceptual reasoning and processing speed in inattentive ADHD, and with poorer working memory in ADHD combined type [38].

For ADHD variables, subtype-related differences were mostly non-significant; both Bechtel et al. [27] and Kral et al. [33] found no major neurocognitive differences across ADHD subtypes, except for a higher number of perseverative errors in the combined type [33]. Regarding medication, Hermann et al. [7] found no cognitive differences in the combined group whether they were treated with stimulants or not. However, Semrud-Clikeman & Wical [4] revealed that stimulants showed benefits in attention performance overall, though not to a level that equalized outcomes between ADHD-only and ADHD+CWE groups, largely owing to the finding that attention was worse at baseline for the combined group. Tsai et al. [42] demonstrated that higher IQ was linked with less probability of ADHD in CWE, describing it as a possible protective factor.

3.3. *Quality assessment and risk of Bias*

Across the eight NOS items, the mean inter-rater agreement (Cohen's κ) was 0.63 (range 0–1), indicating substantial overall agreement. Two items showed lower agreement (κ = 0.30 and 0.00), likely a reflection of the small sample size and limited variability in ratings (e.g., the latter had 100% agreement with all “no” responses). The NOS quality ratings for the 22 studies ranged from 3 to 9, with a mean score of 4.91, indicating a concentration of studies within the moderate risk of bias range. While a few studies demonstrated higher methodological quality (scores of 7 and 9), several studies scored on the lower end, suggesting notable variability in overall study quality (see supplementary Table S1). Most studies applied the same test batteries across groups and adequately selected and defined control groups. The most common sources of potential bias related to limited representativeness in the selection domain of the comorbid group, as well as frequent lack of examiner blinding and incomplete reporting of non-response rates in the exposure domain. Funnel plots were inspected for evidence of publication bias. For analyses with ≥ 10 studies (G and Gc for CWE + ADHD vs CWE only), Egger's test was consulted and was non-significant for both ([G, $z = 0.86$, $p = 0.39$; Gc, $z = 1.85$, $p = 0.06$]).

4. Discussion

The present findings revealed that the group of children with both epilepsy and ADHD displayed poorer overall neuropsychological scores compared to either condition alone with small to moderate effect sizes. Compared to healthy controls, the cognitive deficit in the combined group was, unsurprisingly, markedly larger. This compliments and extends the recent systematic review and narrative synthesis by Leal et al. [19] by revealing the magnitude of cognitive impairment in the combined group compared to CWE and ADHD alone, as well as healthy controls. These findings are consistent with previous meta-analytic reviews of neuropsychological functioning in both CWE [12–14,21] and ADHD [15–16], indicating widespread and heterogenous cognitive outcomes.

The relationship between childhood epilepsy and ADHD is complex and multifaceted, with influences from genetics, disease and treatment variables including specific interactions with localised effects, as well as environmental factors [6]. One interpretation of the present findings at group level is that there may be a dose-response relationship corresponding to amplified cognitive deficits. Children with both conditions exhibit more widespread and/or increased dysfunction of the underlying developmental brain disorder, with heterogeneity interacting with contextual and specific disease variables (e.g., onset of seizures, epilepsy type, frequency of seizures, ADHD subtype etc.). This more parsimonious suggestion is that both disorders share domains of impairment and together, are additive to worsen outcomes. Alternatively, there are also indications for unique contributions from each condition affecting cognitive function in different ways and distinct effects combine to produce a broader, more heterogenous profile of widespread cognitive dysfunction.

While acknowledging the overlapping confidence intervals across the various cognitive domains, there is some evidence from within the neuropsychological profile comparisons to suggest that there are

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perhaps relatively greater contributions arising from certain cognitive constructs from the CWE and ADHD components. Within CHC theory, Executive Functioning (EF) is conceptualised as a conflation of various broad stratum abilities that tend to load more on factors of fluid intelligence, processing speed and retrieval fluency (e.g., corresponding to tasks such as trail making, Stroop, verbal fluency, and tower of Hanoi among others). Compared to healthy controls, effect size point estimates indicate largest cognitive deficit in the combined group to EF domains of fluid reasoning and speed of information processing. On the other hand, there is data to support that inattention symptoms can be directedly related to processing speed and fluid reasoning organised within CHC taxonomy, not otherwise mediated by test session behaviour [45]. The combined group compared to ADHD alone showed relative effects greatest in domains of memory retrieval and speed, which is cited as a consistent feature in CWE compared to controls [46]. Alternatively, the addition of ADHD to CWE (i.e., in the comparison of the combined group to CWE alone) showed relatively higher effect sizes for verbal intelligence and language measures as well as reaction time variability, the latter reflecting a common measure of intraindividual variability and sustained attention that holds well-known clinical utility in the neuropsychological assessment of ADHD [16,47]. Although effect sizes were small for processing efficiency domains of short-term working memory and processing speed in the CWE alone comparison, it is noteworthy that prediction intervals and heterogeneity statistics were lower and non-significant, showing that these effected domains were more predictable and stable across studies.

Across most cognitive domains, there was substantial heterogeneity between studies. Although the meta-analysis revealed significant pooled effect sizes at the group level, the majority of prediction intervals were wide and included zero, indicating that the observed effects in any future study could plausibly be small, large, or even absent. This suggests meaningful variability in cognitive outcomes across samples. In general, confidence and prediction intervals were wider for the healthy control comparison compared to the clinical comparisons, which makes sense given the shared clinical features and impact thereof. At the individual level, this aligns with the broad spectrum of cognitive strengths and vulnerabilities observed clinically—despite an overall increased risk for cognitive deficits. Thus, the substantial between-study heterogeneity would appear to mirror the variability seen at the level of the individual. Taken together, these findings demonstrating significant variability underscore and support the growing movement towards identifying broad cognitive phenotypes within a personalised medicine framework [10].

The evidence from this study lends weight to the clinical relevance of neuropsychological assessment when seizures co-occur with attention and behaviour challenges in children. This is particularly relevant during the diagnostic assessment of ADHD where cognitive testing is common, although not always completed in practice. Given the magnitude of cognitive deficit in the combined group, there is also a need for interventions that can support cognition while addressing seizures and ADHD symptoms. Reviews have shown that stimulant medication is generally tolerable and safe in CWE [8], although there can be caution exercised when treating comorbidity with multiple medication classes in children. As such, medication alternative psychosocial intervention, such as behavioural therapy, cognitive remediation, and/or targeted neuromodulation therapies should also be considered [48]. For example, EEG biofeedback is a non-invasive biobehavioural technique that provides real-time feedback of brain activity to support self-regulation of cognitive, emotional and behavioural functions and there is a body of evidence investigating EEG biofeedback in both ADHD [49] and epilepsy [50]. There were in fact studies that investigated the effectiveness of neurofeedback in children with both CWE and ADHD, conditioning cortical stability through reinforcing sensorimotor rhythm activity and/or feedback on arousal vigilance states [26,51–52]. These studies went beyond intervention for primary symptoms (e.g., to reduce seizures, inattention, hyperactivity etc.), with some evidence showing improvements in cognitive ability such as short-term working memory, and cognitive flexibility speed [26,51]. While the existing research is scarce and limited to small samples and case studies, there is value in further well-controlled studies exploring individualised biologically-driven, behavioural interventions in this at-risk population.

The present study has several limitations. The combining of various childhood epileptic syndromes into a single group limited the ability to examine neuropsychological profiles across epileptic syndromes and may have introduced bias, leading to findings that suggest more widespread and heterogeneous deficits. However, previous reviews using similar methods in specific focal epilepsy syndromes [14,21] and idiopathic generalised epilepsy [13] have shown comparable findings. Additionally, many of the studies that examine CWE and comorbid ADHD tend to combine epilepsy types together. There is the further limitation of collapsing narrow cognitive abilities into broader CHC constructs, which results in a loss of detail regarding specific cognitive functions. There were also several notable CHC broad stratum abilities either under-represented (e.g., G*lr* and G*r*) or absent (i.e. G*v* and G*a*) from the current meta-analysis, given the lack of studies including such measures, that limit a more comprehensive appreciation of the cognitive profile.

Included studies showed wide variability in exclusion criteria related to IQ which introduces bias in representativeness. Where several studies applied no exclusion based on IQ, others varied exclusion from severe intellectual disability to IQ scores below 80. Given the vulnerability of lower cognitive functioning in this combined population, variability in exclusion criteria will likely lead to an underestimation of the impact of cognitive difficulties. Furthermore, many studies failed to report on the status of stimulant medications at the time of cognitive assessment. Given the well-known positive impact of central nervous system stimulants on neuropsychological functioning, especially on tests of attention and executive function [53], this variability in standardization introduces a potential confound in appreciating the influences of cognitive dysfunction in this population.

The present findings support a widespread profile of cognitive dysfunction in children with both epilepsy and ADHD compared to either condition alone or healthy controls, with important consideration to the significant heterogeneity present. Future studies would benefit from more standardised reporting of clinical variables to enable clearer understanding of the complex relationship between neurological and neurobehavioral manifestations. Given the substantial heterogeneity observed at the group level, future research adopting idiographic and contextual approaches may be well suited to elucidate causal factors in the brain-behaviour relationship to inform personalised intervention. In conclusion, CWE who also experience clinically significant issues with attention and behaviour are at an increased risk of neuropsychological dysfunction, and there is a need for further empirical and clinical efforts into targeted interventions for addressing neuropsychological sequelae.

Authorship contribution statement

Steven Wickens: Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing. Persephone Gummersall: Methodology (quality assessment), Writing – review & editing. Trevor Brown: Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the primary author, SW, used ChatGPT to assist with organisation, language refinement to improve readability and to support the drafting of analysis scripts in RStudio. After using this tool/service, the author/s carefully reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: All authors are/**were** employed by Neurocare Group (Melbourne, Australia) who offer psychological services and neuromodulation treatment and training services within their clinics. The authors otherwise declare that they have no conflict of interest or declarations.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.braindev.2026.104533.

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